

Sujal Papalkar

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SUMMARY

Aspiring **AI/ML Engineer** specializing in **Agentic AI, LLM systems, and full-stack AI applications**. Built **RAG pipelines, multi-agent systems, and intelligent platforms** using LangGraph, Gemini API, and FastAPI. Focused on scalable, explainable AI solutions.

TECHNICAL PROJECTS

DocMind – Multimodal RAG Chatbot

FastAPI | Next.js | Gemini | Pinecone | LangChain

- Engineered a **multimodal RAG pipeline** processing PDFs into structured chunks (text, tables, images), handling 40+ page documents
- Implemented async LLM pipelines with parallel execution, reducing latency by **30%**
- Built a **hybrid search** system combining vector search with real-time web search for improved answer relevance
- Developed session-aware architecture with **page-level citations** and multimodal response rendering

AI Code Review Agent – Agentic AI System

Python | FastAPI | LangGraph | SQLite | Next.js | GitHub API | Groq

- Built a **ReAct-based AI system** with up to **3 iterative validation cycles** for improving code fixes
- Designed a **validation pipeline** using syntax checks and verification to ensure reliable code fixes
- Developed a **benchmarking setup (8+ test cases)** to evaluate detection accuracy across issue types
- Implemented **repository memory** to store past reviews and feedback for improved future suggestions

NextStep AI – Intelligent Career Assistant

Spring Boot | Next.js 14 | Gemini 2.0 Flash | PostgreSQL

- Designed **LLM-driven workflows with prompt engineering** to analyze profiles across 5+ categories
- Built a **structured scoring system (0–100)** for consistent and explainable evaluation
- Developed **JD Matcher** using LLM-based ATS analysis to identify match score and missing skills
- Contributed to chatbot module using **context-aware prompts** for personalized career guidance

SKILLS

- Programming & Frameworks:** Python, FastAPI, NumPy, Pandas, Scikit-learn, JavaScript, Next.js, Java
- LLM & Generative AI:** RAG, Prompt Engineering, ReAct, Embeddings, Semantic Search
- AI Frameworks & Tools:** LangChain, LangGraph, Gemini API, Groq API, Hugging Face
- Databases & Systems:** PostgreSQL, MySQL, MongoDB, SQLite, Pinecone
- Core Concepts:** Transformers, NLP, Agentic Workflows, AI Pipelines, Context Engineering
- Dev Tools:** Git, GitHub, Postman, VS Code

EXPERIENCE

Web Developer Intern — TechnoGrowth Software Solutions Pvt. Ltd., Pune

Apr 2025 – Jun 2025

- Optimized backend APIs through query optimization and caching
- Designed and developed REST APIs for production systems
- Collaborated in agile team using Git workflows and code reviews

EDUCATION

Savitribai Phule Pune University

2022 – 2026

Bachelor of Engineering, Computer Engineering | CGPA: **8.33 / 10**

Relevant Coursework: Machine Learning, Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Computer Networks, Software Engineering

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 — Generative AI Professional
- Oracle Cloud Infrastructure 2025 — Foundations Associate